

Appendix C. Expanded Transfer Grids

Following are expanded tables of accuracies from our probing experiments.

Train → eval	L20	L25	L30	L35	L40
instructed → instructed (response)	1.000	1.000	1.000	1.000	1.000
spontaneous → instructed (response)	0.870	0.820	0.980	0.980	0.970
spontaneous → spontaneous (response)	0.918	0.914	0.922	0.929	0.922
instructed (prompt) → spontaneous (response)	0.500	0.500	0.638	0.873	0.851
spontaneous → spontaneous (prompt)	0.575	0.567	0.567	0.541	0.556

Table 5: Binary probe results by layer, seed 42. Bold rows are within-setting results. The best accuracy was achieved at layers 30-40. Spontaneous last prompt token trained probes fail to perform well at all layers.

Train → eval	Seed	Bal. acc.	Decep.	Honest
instructed → instructed (response)	42	1.000	1.000	1.000
	43	1.000	1.000	1.000
	44	1.000	1.000	1.000
spontaneous → instructed (response)	42	0.980	1.000	0.960
	43	0.980	1.000	0.960
	44	0.980	1.000	0.960
spontaneous → spontaneous (response)	42	0.929	0.895	0.963
	43	0.955	0.955	0.955
	44	0.918	0.881	0.955
instructed (prompt) → spontaneous (response)	42	0.873	0.791	0.955
	43	0.914	0.866	0.963
	44	0.892	0.806	0.978
spontaneous → spontaneous (prompt)	42	0.541	0.776	0.306
	43	0.522	0.784	0.261
	44	0.526	0.776	0.276

Table 6: Binary probe results at layer 35 by seed. Binary class(deceptive or honest) accuracies are included along with the balanced accuracy. Collapses can occur in one class such that almost all items are classified as honest or deceptive. Balanced accuracy alone obscures this phenomenon.

Train $\rightarrow$ eval	Accuracy	Fabric.	Omission	Honest
instructed $\rightarrow$ instructed (response)	1.000	1.000	1.000	1.000
spontaneous $\rightarrow$ instructed (response)	0.953	1.000	0.920	0.940
spontaneous $\rightarrow$ spontaneous (response)	0.866	0.925	0.731	0.940
instructed $\rightarrow$ spontaneous (response)	0.512	0.164	0.388	0.985

Table 7: Type-level accuracy results at layer 35 with average accuracy for probes trained on, and applied at, response tokens with seed 42.

Appendix D. Expanded steering tables

Following are expanded tables of steering runs from our experiments. All runs steer at all tokens, layers 19-32. Prompt (last prompt token) derived and response derived refer to where the directions were extracted from.

Mech.	Direction, vector	Base	-1.5	-1	-0.5
fabrication	instructed, prompt-derived	6.2 (100)	3.7 (100)	1.3 (100)	1.0 (100)
	instructed, response-derived	6.2 (100)	10.9 (93)	2.0 (100)	1.1 (100)
	spontaneous, prompt-derived	6.2 (100)	9.1 (100)	3.6 (100)	1.4 (100)
	spontaneous, response-derived	6.2 (100)	35.8 (80)	20.4 (99)	2.3 (100)
random (6 draws)		—	3.1–11.8	4.1–10.5	4.5–7.0
omission	instructed, prompt-derived	49.7 (100)	14.2 (97)	17.4 (99)	25.6 (99)
	instructed, response-derived	49.7 (100)	61.1 (92)	26.3 (99)	31.8 (100)
	spontaneous, prompt-derived	49.7 (100)	23.3 (99)	14.0 (100)	25.8 (100)
	spontaneous, response-derived	49.7 (100)	75.4 (56)	55.2 (82)	37.4 (99)
random (6 draws)		—	40.2–63.8	44.7–59.9	47.4–55.1

Table 8: Results for steering on spontaneous prompts with negative coefficients (suppression) at layers 19-32. Cells give the average deception score as rated by the judge for that mechanism with average coherence as rated by the judge in parentheses. Base is results for the baseline unsteered run. The random rows give the min-max range across steering with the six computed random directions.

Mech.	Direction, vector	Base	+0.5	+1	+1.5
fabrication	instructed, prompt-derived	6.2 (100)	61.4 (99)	84.8 (97)	78.7 (95)
	instructed, response-derived	6.2 (100)	37.0 (99)	41.0 (98)	45.6 (91)
	spontaneous, prompt-derived	6.2 (100)	36.2 (98)	45.9 (93)	29.0 (68)
	spontaneous, response-derived	6.2 (100)	28.5 (98)	40.8 (95)	47.8 (86)
random (6 draws)		—	4.2–7.1	3.9–7.3	3.4–7.4
omission	instructed, prompt-derived	49.7 (100)	73.2 (98)	75.5 (96)	80.6 (91)
	instructed, response-derived	49.7 (100)	70.2 (99)	70.6 (98)	86.1 (80)
	spontaneous, prompt-derived	49.7 (100)	76.8 (98)	87.8 (98)	97.1 (95)
	spontaneous, response-derived	49.7 (100)	57.0 (99)	77.8 (91)	86.3 (76)
random (6 draws)		—	43.7–51.5	37.1–51.7	32.7–51.6

Table 9: Results for steering on spontaneous prompts with positive coefficients(induction) at layers 19-32. Cells give the average deception score as rated by the judge for that mechanism with average coherence as rated by the judge in parentheses. Base refers to results for the baseline unsteered run. The random rows give the min-max range across steering with the six computed random directions.

Mech.	Direction, vector	Base	−1.5	−1	−0.5
fabrication	instructed, prompt-derived	91.0 (90)	6.1 (93)	6.9 (94)	66.5 (90)
	instructed, response-derived	91.0 (90)	18.6 (85)	19.1 (90)	75.7 (90)
	spontaneous, prompt-derived	91.0 (90)	43.0 (91)	37.8 (92)	77.3 (90)
	spontaneous, response-derived	91.0 (90)	49.2 (66)	60.0 (89)	81.8 (90)
random (6 draws)		—	76.8–92.5	82.7–92.3	89.0–92.8
omission	instructed, prompt-derived	79.9 (91)	20.3 (92)	14.3 (93)	67.6 (91)
	instructed, response-derived	79.9 (91)	67.3 (83)	58.0 (90)	81.4 (92)
	spontaneous, prompt-derived	79.9 (91)	23.4 (96)	40.2 (91)	78.2 (91)
	spontaneous, response-derived	79.9 (91)	88.8 (41)	95.2 (48)	79.7 (91)
random (6 draws)		—	78.5–80.9	79.3–81.0	79.7–80.5

Table 10: Results for steering on instructed prompts with negative coefficients(suppression) at layers 19-32. In each prompt, the model received an instruction to answer deceptively through the particular mechanism. Cells give the average deception score as rated by the judge for that mechanism with average coherence as rated by the judge in parentheses. Base is results for the baseline unsteered run. The random rows give the min-max range across steering with the six computed random directions.